METRIC
MIL-PRF-53007B(AT)
2 January 1996
SUPERSEDING
MIL-R-53007A(ME)
20 May 1986

PERFORMANCE SPECIFICATION

ROLLER KIT, ROLLER, MINE CLEARING: TRACK-WIDTH, TANK-MOUN'TED

This specifiCAtion is approved for use within U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines the requirements for a roller kit used in the mine clearing roller system kit (M60 series tank).

2. APPLICABLE DOCUMENTS

2.1 . The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommendATION for additional information or as examples. While every effort has been made to ensure the completeness of this list document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-EIBLUE, W wren, Ml 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-M-53008 - Mounting Kit, Roller, Mine Clearing: Track Width, Tank Mounted.

(Unless othenvise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government drawings. The following drawings form a part of this document to the extent specified herein. Unless othenvise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

12369000 - Chemical Agent Resistant Coatings (CARC) Paint Systems Index.
13222E0327 - Roller Kit, Mine Clearing Roller.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Wanen, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted).

(Application for copies should be addressed to American Society for Quality Control, 611 East Wisconsin Avenue, Milwaukee, WI 53202-4606.)

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1 - Structural Welding Code Steel (DoD Adopted).

(Application for copies should be addressed to the American Welding Society, Inc., 550 N. W. LeJune Road Miami, FL 33126.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC - Boiler and Pressure Vessel Code.

(Application for copies should be Addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS478 - Identification Marking Methods (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. Requirements for the roller kit shall be in accordance with Drawing 13222E0327 and as specified herein.

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3.2. Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the kit. Any data, such as, shop drawings, layouts, flow sheets and processing procedures prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request, for inspection by the contracting officer or designated representative (see 4.6.1).

3.3. First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3). Any changes or deviations of the roller kit from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish roller kits conforming to this specification.

3.4 Materials. Materials shall be as specified on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of the specification (see 3.4.4 and 4.6.1).

3.4.1 Material deterioration prevention and control. The roller kit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that maybe encountered in any of the applicable operation and storage environments to which the item maybe exposed (see 4.6.1).

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion (see 4.6.1).

3.4.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative (see 4.6.1).

3.4.4. Recycled, recovered or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs (see 6.5. 1). Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the roiler kit (see 4.6.1).

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3.5 Safety. Nonfunctional excessive length of fastening devices shall be avoided.

3.6 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts and without selection for fit or performance.

3.7 Treatment and painting. Unless otherwise specified (see 6.2 and 6.6), all parts requiring treatment and/or paint topcoat shall be treated and/or topcoated with a paint conforming to Drawing 12369000 for color green 383.

3.8 Marking. Identification and marking shall be permanent and legible in accordance with SAE AS478.

3.9 Workmanship. Each kit shall be clean and free from sand dirt, fins, pits, scale, rust, burrs, sharp edges, projecting points, cracks, poor welds or any other defect that could impair its function.

3.9.1 Metal fabrication. Metal used in fabrication shall be free from rust and corrosion. FIame cutting, using tips suitable for the thickness of the steel, maybe employed instead of shearing and sawing. Precaution shall be taken to avoid overheating. Proper cooling rates should be maintained to prevent adverse effects to the properties of the metal. External surfaces shall be free from burrs, sharp edges and corners, except when sharp edges or comers are required or where they are not detrimental to safety.

3.9.2. Welding. Surface preparation and welding practices shall conform to AWS D1.1. The welding process used in fabrication of the kit shall be as specified in applicable drawings. No deviation from these requirements are permissible without prior Government approval.

3.9.2.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer upon request with certification that the welder has passed qualification test as prescribed by either AWS D1. 1 (Qualification) or ASME Boiler and Pressure Vessel Code (Welding Qualifications) for the type of welding operations to be performed and that the qualification is effective as defined by the applicable code (see 6.7).

3.9.2.2 Inspection. Inspect all welds as per drawing. No cracks are permissible. Workmanship specimens required by the applicable specifications shall be approved by representatives of the contracting activity.

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3.10 Installation test. The installation shall be conducted in accordance with 4.3.3.2.

3.11 Operational test. The operational test shall be conducted in accordance with 4.3.3.3.

3.12 Lubricants. Lubricants shall be as specified in the contract or order (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).
c. Inspection comparison (see 4.5).

4.2 First article inspection. One or more first produced kits from the first 10 units being fabricated by production tooling will be selected by the Government for examination and tests (see 6.3). These kits will be examined for the characterisitics listed in table I and subjected to the test specified in 4.3.3 to determine conformance to the requirements of this specification. The inspection will be performed by the contractor at the contractors location and witnessed by Government inspectors. Acceptance of a first article shall not exclude the remaining kits from the conformance inspection and acceptance provisions specified in 4.3. In addition to any test specified as part of the first article test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test. Failure of such additional tests shall have the same effect as failure of those tests specified as first article tests.

TABLE I. Examination schedule.

Category	Characteristics	Method of examination	First article	Conform-ance
Major:				
101	Dimensions are not as specified (see 3.2).	SIE 1/	X	X
102	Parts and components are not complete (see 3.2).	Visual	X	X
103	Material is not as specified (see 3.4).	Visual	X	X
104	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment (see 3.4.1).	Visual	X	X
105	Dissimilar metals are not effectively insulated from each other (see 3.4.2).	Visual	X	X

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TABLE I. Examination schedule - Continued

Category	Characteristics	Method of examination	First article	Conform-ance
106	Contractor does not have documentation available for identification of material, material finishes or treatments (see 3.4.3).	Visual	x	x
107	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the roller kit (see 3.4.4).	Visual	x	x
108	Safety is not as specified (see 3.5).	Visual	x	x
109	Interference noted (see 3.10 and 3.11).	Visual	x	x
110	Treatment and painting not as specified (see 3.7).	Visual	x	x
111	Paint color not as specified (see 3.7).	Visual	x	x
112	Camouflage patterns not as specified (see 3.7).	visual	x	x
113	Workmanship is not as specified (see 3.9).	visuaI	x	x
114	Welding certification not as specified (see 3.9.2.1).	Visual	x	x
115	Lubricants not as specified (see 3.12).	Visual	x	x
116	Welds not as specified (see 3.9.2).	Visual	x	x
117	Interchangeability not as specified (see 3.6).	Visual	x	x
Minor				
201	Identification marking is not as specified (see 3.8).	Visual	x	x

4.2.1 Inspection failure. Failure of a first article kit to meet any requirement specified herein during, and as a result of; examination and tests specified in 4.2 and 4.3 shall be cause for refusal by the Government to continue acceptance of production kits until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shali be accomplished by the contractor at no cost to the Government. Any deficiencies found as a result of the first article inspection will be considered evidence that all kits accepted prior to the completion of first article inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.3 Conformance inspection.

4.3. Sampling. Sampling for examination and test shall bc in accordance with ASQC Z1.4.

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4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined for the major and minor defects specified in table I. Failure of any examination shall be cause for rejection.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be subjected to the following tests, plus any other tests specified by the contracting activity (see 6.2), to comply with requirement of 3.3. Failure of any test shall because for rejection.

4.3.3.2 Installation test. Both a left and right mine roller assembly shall be installed on an M60 tank equipped with a mine clearing roller mounting kit in accordance with MIL-M-53008. The roller will be installed using the winch and other mounting equipment included with the mine roller mounting kit The mine roller shall be released and remounted at least twice using the hydraulic release mechanism and using the manual release mechanism. During the mounting and release of the roller, the operation shall be observed and any interference of the parts noted (see 3.10).

4.3.3.3 Operational test. Following installation of the mine roller, an operational test shall be conducted consisting of 2 miles of level cross-country operation. During the operational test, both left and right turns and forward atid reverse operations shall be demonstrated. Following the operational test, the mine roller shall be released and remounted at least twice using the hydraulic release mechanism, and twice using the manual release mechanism. During the mounting and release of the roller, the operation shall be observed and any interference of the parts noted. Initial production operational tests for a mine roller kit and mine roller mounting kit may be ran concurrently (see 3.11).

4.4 Inspection schedule.

4.4.1 Examination

4.4.1.1 Examination conditions. Unless othervise specified in this specification, conformance examination shall be performed by the contnctor at the contractor’s location and witnessed by Government inspectors.

4.4.1.2 Examination procedures (first article). One or more first article kits, selected at random, shall be examined for conformance to dimensions specified on the drawings.

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4.5 Inspection comparison. The Government may select kits at any time during the initial production period and subject them to the examinations specified in 4.3.2 and the test specified in 4.3.3 to determine conformance to the requirements of this specification. The inspection will be performed by the contractor at the contractors location and witnessed by Government inspectors on kits selected at random from those which have been accepted by the Government, but will not include the previously inspected first article model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison. Failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison kit to meet any requirement specified herein during and as a result of the examination and tests specified shall be cause for rejection of the inspection comparison kit and shall be cause for refusal by the Government to continue acceptance of first article until evidence has been provided by the contractor, at no cost to the Government that all deficiencies have been corrected. Any deficiencies found as a result of the inspection comparison will be considered evidence that all kits accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contracting officer.

4.6 Inspection failure.

4.6.1 Materials. Conformance to 3.2 and 3.4 thru 3.4.4 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shaIl include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contractor order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnl need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which maybe helpful, but is not mandatory.)

6.1 Intended use. The kit mtains a left and right assembly to be attached to the front of a modified M60 series tank which will neutralize single pulse pressure mines in front of the tank's track and tilt rod mines in the tank belly area.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited solution and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. If fiarst article is required (see 3.3).
d. Color required when other than specified (see 3.7 and 6.6).
e. Lubricant requirements (see 3.12).
f. Additional test requirements (see 4.3.3.1).
g. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the items should be a first production model. The first article should consist of one or more kits. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 Subiect term (key word) listing.

Belly area
Left assembly
Modified M60 series
Neutralize
Right assembly
Single puke pressure
Tilt rod

6.5 Definitions.

6.5.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6 5.2).

6.5.2 Solid waste. "Solid waste" (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatnmt plant, or air pollution control facility; and (b) other discarded

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material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. lt does not include solid or dissolved materials in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits uncier sections 402 of the Clean Water Act, (33 U. S .C. 1342 et seq.), or source, special nuclear, or byproduct. material as defined by the Atomic Energy Act of 1954042 U.S.C.201 1 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.6 Government-furnished property. The contracting officer should arrange to furnish the following property as required:

Item No.	Description	Iidentification	Quantity
1	Pattern drawings	Camouflage patterns	As required

6.7 Welders qualification. Subject to approval by the Government, the contractor's standard welder qualification may be substituted, provided that the contractor's procedure is equivalent to the AWS and ASME codes (see 3.9.2.1).

Custodian:
Army - AT

Preparing activity:
Army-AT

(Project 1095-0292)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-53007B(AT)	960102

3. DOCUMENT TITLE
Roller Kit, Roller, Mine Clearing: Track-Width, Track-Mounted

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (810) 574-8745	(2) AUTOVON 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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